

TRIGGERS:

A trigger is a special type of procedure that will be used to provide restriction on the tables when a language event is executed. SQL server includes two types of triggers:

- DML Triggers
- DDL Triggers

DML Triggers: DML triggers execute when the user tries to modify or change data through data manipulation language events. These are Inserting, Update and Delete statements on the table.

DML triggers can be used to enforce business rules and data integrity. With the help of a DML trigger we can enforce integrity which cannot be done with constraints.

Syntax:

```
Create Trigger <Trigger Name>
on <table Name> For [Insert, Update,Delete]
AS
```

```
Begin
<Statements>
End
```

Ex: A trigger that will convert the dname and location into upper case when the user inserts in lowercase.

```
create trigger per_trg on person after insert as
begin
declare @pid int,@pname varchar(50),@loc varchar(50) select
@pid=pid,@pname=pname,@loc=loc from inserted
update person set pname=upper(@pname),loc=upper(@loc) where pid=@pid end
```

Ex: Create a trigger to restrict DML operations on the table

```
create trigger nnn on person
for insert,update,delete as
begin
print 'DML Operation are Not Allowed' rollback transaction
end
```

EX: A Trigger that will restrict the operations to be performed before 9 A.M and after 5 P.M

```
CREATE TRIGGER EMP_TRG
ON EMP AFTER INSERT, UPDATE, DELETE AS
BEGIN
DECLARE @DT INT
SET @DT=DATENAME(HH, GETDATE()) IF @DT NOT BETWEEN 9 AND 16 BEGIN
ROLLBACK
RAISERROR('CANNOT PERFORM DML OPERATIONS NOW', 15, 1) END
END
```

-After the trigger is created try to perform any DML Operations on the EMP table before 9 A.M and after 5 P.M the Trigger will fire and restrict the operations.

EX: A program which will restrict the Delete operation if the Job of the person is Manager.

```
ALTER TRIGGER EMP_DELETE_TRG ON EMP AFTER DELETE
AS
BEGIN
DECLARE @JOB VARCHAR(50) SELECT @JOB=JOB FROM DELETED IF @JOB='MANAGER'
BEGIN ROLLBACK
RAISERROR('CANNOT DELETE MANAGER FROM THE TABLE', 15,
1)
END END
```

-To test the following Trigger execute the following statement: DELETE FROM EMP WHERE EMPNO=1002

EX: A Trigger which will restrict to update the Salary of the Employee if the New Salary is less than the Old Salary.

```
CREATE TRIGGER EMP_UPDATE_TRG ON EMP AFTER UPDATE
AS BEGIN
DECLARE @OLDSAL MONEY

DECLARE @NEWSAL MONEY
SELECT @OLDSAL=SAL FROM DELETED SELECT @NEWSAL=SAL FROM INSERTED IF @OLDSAL > @NEWSAL
BEGIN ROLLBACK
RAISERROR('NEW SAL CANNOT BE LESS THAN OLD SAL', 15, 1) END
END
```

Dropping DML Triggers:

Syntax: Drop <Trigger> <Trigger Name>

EX: Drop Trigger rest drop

DDL Triggers:

DDL triggers fire in response to a data definition language event like create, Alter, drop etc. A DDL trigger is a special type of procedure that executes in response to a server scoped or database scoped events.

Syntax:

```
Create Trigger <Trigger Name>
on database
after <Event type>
As
Begin
<Statements>
End
```

Ex: Write a trigger which restricts dropping of a table from the database.

```
create trigger restdrop on database after drop_table
as begin rollback
raiserror('Can not drop table under this database',1,1) end
```

Ex2: Write a trigger which restricts Creating of a table from the database.

```
create trigger restcret on database after
create_table
as
begin rollback
raiserror('Can not create table under this database',1,1) end
```

Ex3: Write a trigger which restricts Alter of a table from the database. create

```
trigger restalt on database after Alter_table
as
begin rollback
raiserror('Can not Alter table under this database',1,1);end
```

Dropping DDL Triggers:

Syntax: Drop <Trigger> <Trigger Name> on Database

EX: Drop Trigger rest drop on database

Magic Tables: SQL Server allows you to define a Magic Table.

Magic Tables are invisible tables or virtual tables.

You can see them only with the help Triggers in SQL Server.

☐ Magic Tables are those tables which allow you to hold inserted, deleted and updated values during insert delete and update DML operations on a table in SQL Server.

☐ Basically there are two types of magic table in SQL server namely inserted and deleted magic tables

update can be performed with help of these twos.

Generally we cannot see these two table,
we can only see it with the help Trigger's in SQL server.

Inserted Magic Table: Whenever you insert a record on that table, that record will be shown in the INSERTED Magic Table. Now creating a trigger to see the data in Inserted Magic table.

Example:

```
Create TRIGGER Trigger_ForInsertmagic ON Employee
FOR INSERT AS
Begin
SELECT * FROM INSERTED
End
```

Now insert a new record in Employee table to see data within Inserted virtual tables.

Insert into Employee values (12, 'Rahul', 25000, "HYD") SELECT * FROM Employee

Exception Handling: We handle errors of a program both in a programming language as well as databases also. whereas handling an error in a programming language needs stopping the abnormal termination and allowing the statements which are not related with the error to execute where as handling as error in sqlserver means stopping the execution of statements which are related with the error

Handling Errors In SQL Server: From sqlserver 2005 we are provided with a structure error handling mechanism with the help of TRY and CATCH blocks which should be used as following,

```
Begin Try
<Statements> End Try Begin Catch
<Statements> End Catch
```

Ex: A procedure for dividing two numbers

```
CREATE PROCEDURE PDIV(@X INT,@Y INT) AS
BEGIN
DECLARE @Z INT BEGIN TRY
SET @Z=@X/@Y
PRINT 'THE RESULT IS:-'+CAST(@Z AS CHAR) END TRY
BEGIN CATCH
PRINT ERROR_Message() END CATCH
END
```

Exec PDIV 100,5

Error Message(): It is used to display the information about the error occurred.

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